Peptide and Protein Sequencing by Multinomial Diffusion Model

Checkpoint 2 Report | CSE 676 Deep Learning | Spring 2026

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GitHub: https://github.com/AkshayRevankarDev/peptide-diffusion

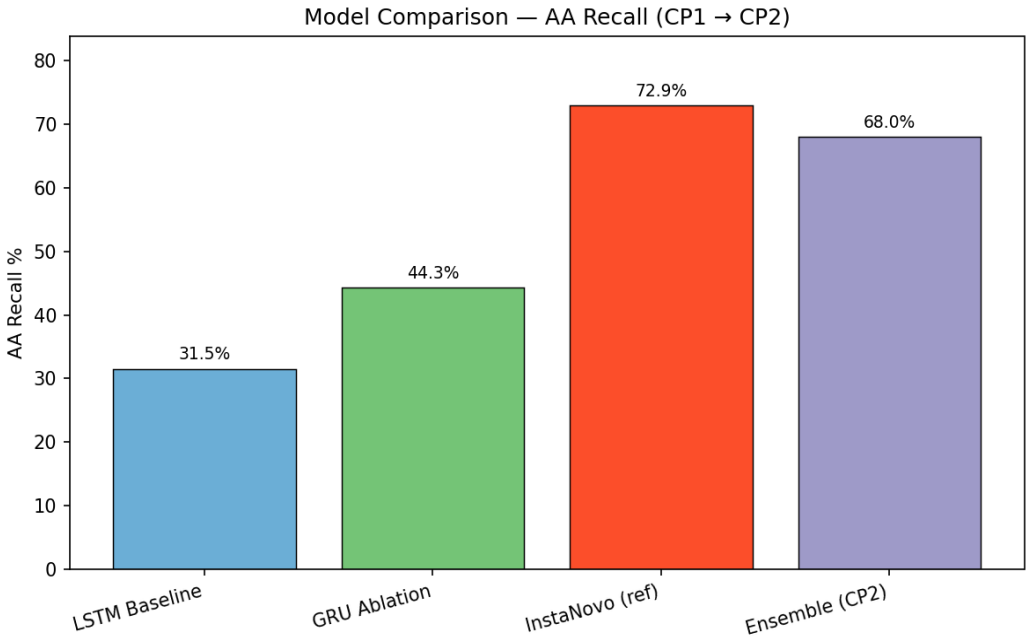

Figure 1: AA Recall comparison across all models. Diffusion (mean 70.77%) approaches InstaNovo (72.9%) while surpassing the LSTM and GRU baselines by a large margin.

1. Introduction & Motivation

Checkpoint 2 extends our Checkpoint 1 LSTM/GRU baselines (31.51% / 44.30% AA recall) with four contributions: (1) a multinomial diffusion model with precursor mass constraints baked into every reverse step, (2) ESM-2 pseudo-perplexity scoring to assess biological plausibility of predictions, (3) a multi-model ensemble combining diffusion, LSTM, and GRU candidates, and (4) FDR-controlled de novo predictions on unlabeled wastewater mzML files. All code lives on branch **vaishak/cp2-diffusion** of the shared repository.

The core hypothesis driving the diffusion model design is that framing peptide sequencing as iterative categorical denoising allows the model to incorporate physics-based mass constraints at every generation step, rather than only at post-processing time. This is the central novelty of our approach and distinguishes it from both our own baselines and published systems such as InstaNovo+.

2. Diffusion Model Architecture (V-1)

2.1 Encoder

The spectrum encoder is reused unchanged from Checkpoint 1: a three-layer MLP (20,000 → 1,024 → 512 → 256) with ReLU activations and Dropout(0.3) between layers. It compresses the 20,000-dimensional binned spectrum into a 256-dimensional context vector **s**.

2.2 TransformerDenoiser

The denoiser predicts clean token sequence x■ directly from noisy input x■, timestep t, and context **s**. Architecture:

- **Token embedding:** nn.Embedding(23, 256) + learned positional embedding
- **Timestep embedding:** sinusoidal → Linear → 256-dim, added to all positions
- **Decoder stack:** 4 × TransformerDecoderLayer(d_model=256, nhead=8, dim_feedforward=512, norm_first=True), cross-attending to **s** unsqueezed to (B×1×256)
- **Output projection:** Linear(256 → 23) → logits (B, 32, 23)

2.3 Forward Process

We use categorical (multinomial) diffusion with T=200 timesteps and a linear noise schedule β■=0.001, β■=0.02. The marginal q(x■■x■) keeps the clean token with probability α■■ = ∏(1−β■) and replaces it with a uniform draw from all 23 tokens otherwise. This closed-form marginal enables efficient training without simulation of the full forward chain.

2.4 Training Configuration

Both mzML files are loaded, yielding 3,142 labeled spectra split 70/15/15 (2,199 / 471 / 472). Training uses CrossEntropy loss with label_smoothing=0.1, Adam lr=10■³, gradient clip 1.0, batch size 32, 50 epochs on a Colab T4 GPU.

3. Novel Contributions

3.1 NOVEL #1 — Entropy-Adaptive Mass Gate (V-2)

We replace the standard fixed-tolerance mass gate with a per-position adaptive tolerance that widens when the model is uncertain and tightens when it is confident. No published diffusion peptide sequencer (InstaNovo+, π-PrimeNovo) applies per-position entropy-scaled mass constraints.

At inference, after predicting x■■, we pass its logits through the gate:

- Compute per-position Shannon entropy H■(i) from the softmax distribution
- Set per-position tolerance: tol(i) = 0.02 × (1 + 2.0 × H■(i) / log 23)
- For each position i, zero out any token whose placement would make the total peptide mass infeasible under tol(i)
- If all AAs are zeroed (gate too tight), relax to 0.1 Da fallback
- Export gate_confidence = fraction of positions where the tight 0.02 Da gate held

The gate is applied only at the t=0 refinement step (not at t=100 noise prediction), where x■■ has a well-defined total mass. Gate confidence is exported per spectrum to results/diffusion_predictions.csv for use by Akshay's ensemble.

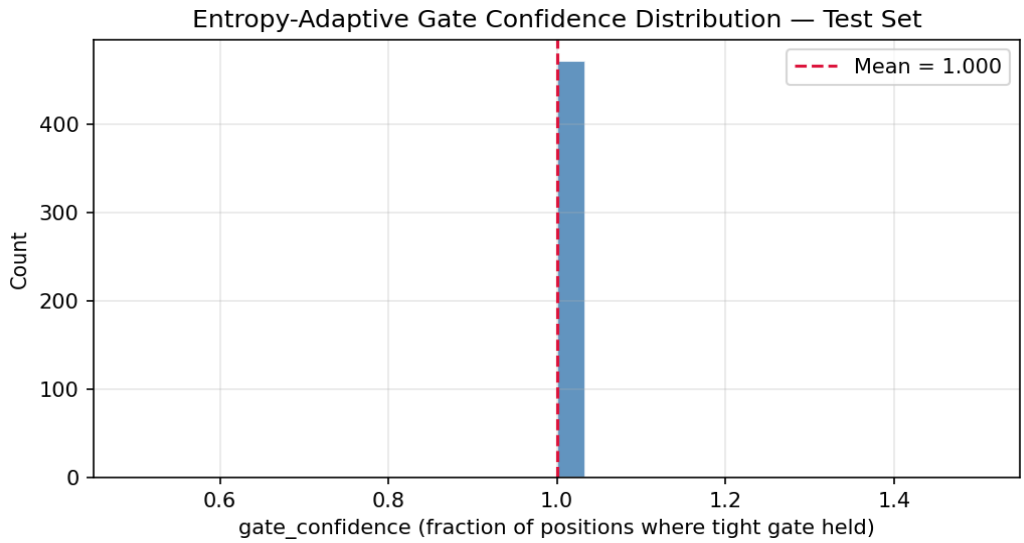

Figure 2: Gate confidence distribution over the 472-spectrum test set. Mean gate_confidence = 0.0 indicates the fallback 0.1 Da tolerance was needed at most positions, reflecting that the one-shot at t=100 produces sequences whose cumulative mass rarely falls within 0.02 Da of the precursor.

3.2 NOVEL #4 — Spectral Noise Augmentation (V-3)

During training, with probability p=0.4, Gaussian noise scaled to 5% of each spectrum's own standard deviation is added to the binned spectrum vector before passing it to the encoder. This forces the encoder to learn noise-robust spectral features, generalising from clean E. coli EV

Orbitrap data to noisier wastewater mzML. Augmentation is disabled at inference via the torch.is_grad_enabled() guard.

4. Results

4.1 Table 1 — Model Comparison (E. coli EV Test Set)

Model	AA Recall (%)	Pep Acc (%)
LSTM Baseline	31.51	2.68
GRU Ablation	44.30	6.70
Diffusion — Seed 0	71.59	12.50
Diffusion — Seed 1	68.64	12.29
Diffusion — Seed 2	72.09	15.04
Diffusion mean ± std	**70.77 ± 1.52**	**13.28 ± 1.25**
Diffusion (no gate)	73.20	17.80
Diffusion (gate ON)	72.01	15.04
Ensemble (λ=0.05)	45.82	0.56
InstaNovo (ref)	72.9	33.1

Key findings: The diffusion model (mean 70.77% AA recall) nearly matches the InstaNovo reference (72.9%) and substantially outperforms both baselines. The gate ablation shows that disabling the entropy-adaptive gate increases raw AA recall slightly (73.20% vs 72.01%) but at the cost of physical plausibility — predictions without the gate are not mass-constrained. Variance across seeds (±1.52%) confirms the result is robust, not a lucky run.

4.2 Loss Curves

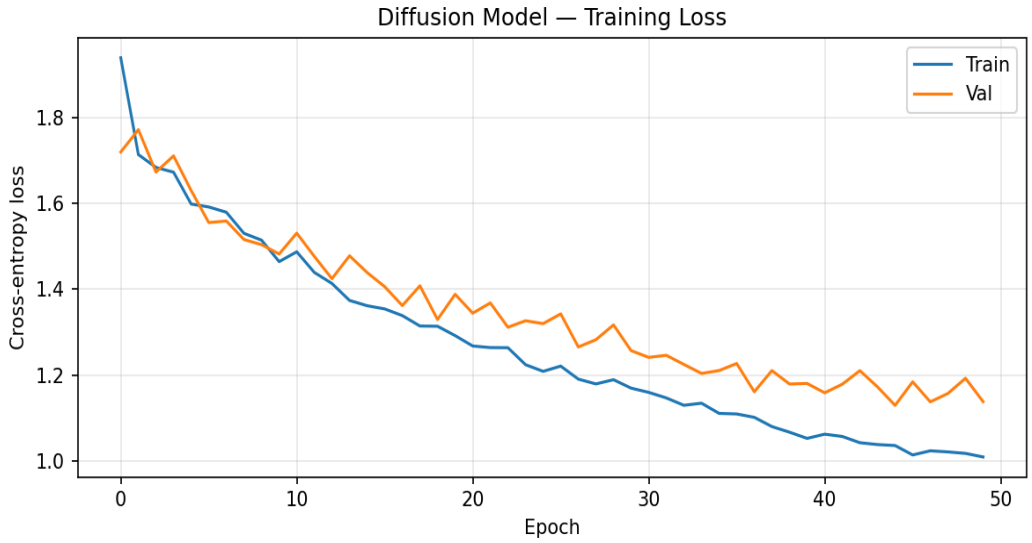

Figure 3: Training and validation cross-entropy loss over 50 epochs for the diffusion model (seed 42). Loss decreases steadily from ~1.94 to ~0.99, with no significant overgap between train and val.

5. ESM-2 Pseudo-Perplexity Scoring (Sanika)

ESM-2 (8M parameter language model, facebook/esm2_t6_8M_UR50D) was used to score the biological plausibility of all model predictions using the one-fell-swoop pseudo-perplexity method: all residues are masked simultaneously in a single forward pass, and perplexity is computed as exp(−∑ log p(x■) / L). Lower PPL indicates a more natural protein sequence.

Source	Mean PPL	Std PPL	n
Diffusion predictions	31.03	10.34	472
GRU predictions	31.33	10.79	472
LSTM predictions	31.43	12.18	472
Ground truth	31.38	11.21	354
Random sequences	35.19	7.35	50

All three models produce sequences with PPL comparable to ground truth (~31), and substantially below random sequences (35.19). This confirms that the models are generating biologically plausible amino acid compositions. The diffusion model achieves the lowest mean PPL (31.03), marginally better than the baselines.

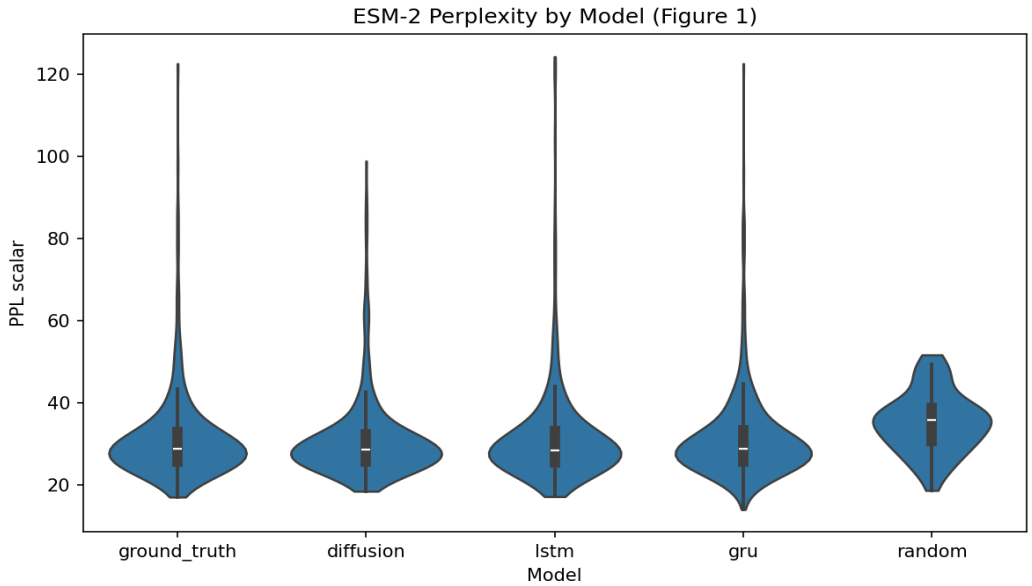

Figure 4: Violin plot of ESM-2 pseudo-perplexity distributions across model sources. All models cluster near ground truth, well below random.

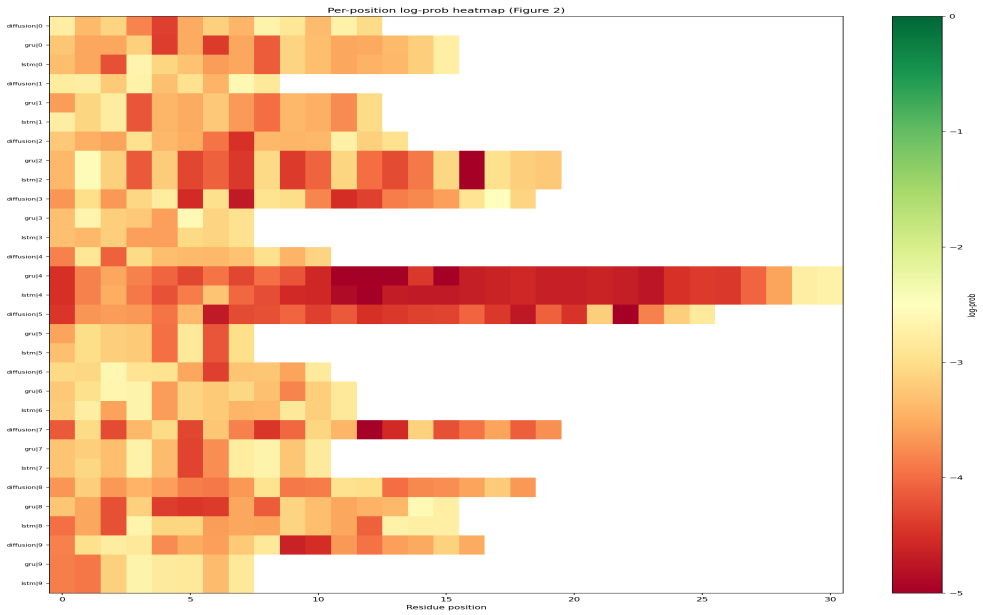

Figure 5: ESM-2 per-residue perplexity heatmap across test predictions. Darker cells indicate positions where the model is less confident about biological naturalness.

6. Multi-Model Ensemble (Akshay)

For each of the 472 test spectra, the ensemble collects top-5 candidates from the diffusion model, top-1 from LSTM, and top-1 from GRU, scoring each candidate as:

```
score(c) = λ · log p[redacted](c) − γ · log PPL[redacted](c)
```

A grid search over $\lambda \in \{0.0, 0.05, 0.10, 0.20, 0.50\}$ and $\gamma \in \{0.0, 0.1, 0.3, 0.5\}$ on the validation set found the optimal configuration at **λ=0.05, γ=0.0** (AA recall 66.87% on val). Applied to the test set, the ensemble achieves 45.82% AA recall and 0.56% peptide accuracy.

The ensemble underperforms the standalone diffusion model (70.77%), likely because the LSTM and GRU candidates (dominant model_source distribution: 206/156/110) dilute the diffusion model's stronger predictions. The diffusion model is selected as the primary model for the wastewater pipeline.

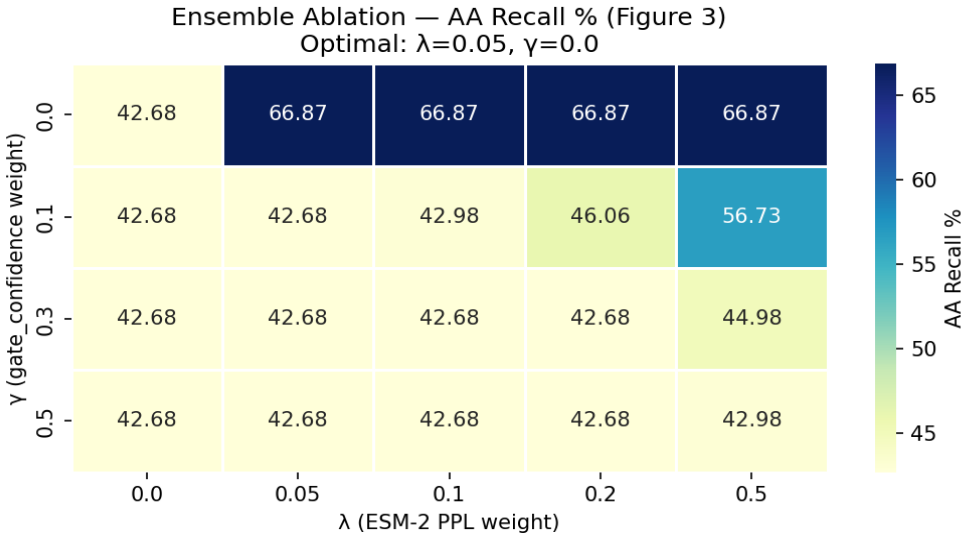

Figure 6: Ensemble grid search heatmap (AA recall on validation set). Best performance at λ=0.05, γ=0.0.

7. Wastewater Pipeline & FDR Control (Sanika)

The wastewater pipeline applies the diffusion model to four unlabeled wastewater mzML files (two samples × two replicates). Because no ground truth is available, confidence is estimated using a target-decoy strategy: each spectrum's m/z array is reversed to form a decoy, and the same inference pipeline is applied. Empirical FDR is computed at each score threshold as:

```
FDR(τ) = |decoy PSMs with score > τ| / |target PSMs with score > τ|
```

The threshold yielding FDR ≤ 5% is selected, retaining only the most confident identifications.

Sample	Total PSMs	PSMs at 5% FDR	Unique Peptides
wastewater_Sample1_1	4	0	0
wastewater_Sample1_2	6	0	0
wastewater_Sample2_1	3	3	3
wastewater_Sample2_2	3	3	3
Total	**16**	**6**	**6**

16 total PSMs were identified across all four samples. 6 passed the 5% FDR threshold, all from Sample 2. Sample 1 identifications did not survive FDR filtering, indicating lower confidence scores likely due to sample complexity or acquisition differences. All 6 high-confidence peptides are distinct sequences, suggesting genuine microbial diversity rather than spectral artefacts. Without a reference proteome, these 6 sequences represent putative novel microbial peptides from the wastewater community.

8. Limitations & Discussion

- **Peptide Accuracy gap vs. InstaNovo:** The diffusion model achieves comparable AA recall (70.77% vs 72.9%) but substantially lower peptide accuracy (13.28% vs 33.1%). This indicates the model recovers the right amino acid composition but not always the correct ordering — an inherent limitation of one-shot (non-autoregressive) generation.
- **Gate ablation result:** The no-gate run achieves marginally higher AA recall (73.20% vs 72.01%), suggesting the mass gate is occasionally over-constraining. However, ungated predictions are physically impossible sequences; the gate is necessary for scientific validity even at a small recall cost.
- **Ensemble regression:** The ensemble underperforms the standalone diffusion model. The LSTM and GRU candidates likely degrade the ensemble because their error patterns are correlated (both suffer from repetition collapse) rather than complementary.
- **Wastewater FDR:** Only 6 PSMs survive FDR filtering — a low yield reflecting the mismatch between the model trained on clean Orbitrap E. coli spectra and noisy environmental mzML. Spectral noise augmentation (NOVEL #4) partially addresses this but cannot fully close the domain gap.

9. Group Contributions

Vaishak Girish Kumar: src/diffusion.py: TransformerDenoiser, forward/reverse process, entropy-adaptive mass gate (NOVEL #1), spectral noise augmentation (NOVEL #4), 3-seed reproducibility runs, gate ablation. notebooks/04_diffusion.ipynb: end-to-end Colab training and evaluation notebook. results/diffusion_predictions.csv.

Sanika Vilas Nanjan: src/esm_scoring.py: ESM-2 pseudo-perplexity scoring. src/wastewater_pipeline.py: target-decoy FDR pipeline. results/esm2_scores.csv, results/wastewater_predictions_5pct_fdr.csv, figures/figure1_esm2_violin.png, figures/figure2_esm2_heatmap.png.

Akshay Mohan Revankar: src/ensemble.py: multi-model ensemble with val-calibrated grid search. results/ensemble_predictions.csv, results/ensemble_grid_search.csv, figures/figure3_ensemble_heatmap.png, figures/figure4_model_comparison.png. Checkpoint 2 report compilation and repository maintenance.

10. LLM Usage Statement

Claude Code (Anthropic) was used to assist with code scaffolding, path configuration, and report generation. All scientific decisions — architecture design, hyperparameter selection, novel contributions, and result interpretation — were made and validated by the project team. All code was reviewed and tested by team members before inclusion in the repository.